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Application of Power Electronics for Renewable Generation - Assignment 1
Report

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1 Task 1.1 - Basic VSC Model with PLL and Current Loop Control

1.1 Introduction

A Voltage Source Converter (VSC) enables bidirectional power flow between AC and DC systems using controllable switches (e.g., IGBTs). Its core components include a DC link capacitor for voltage stabilization, switching devices for PWM-based AC voltage generation, and an output filter (L/LC) to reduce harmonics. For grid-connected operation, synchronization and precise current regulation are achieved via a Phase-Locked Loop (PLL) and current loop control.

The PLL synchronizes the VSC with the grid by tracking the phase and frequency of the voltage of the grid. It uses a phase detector to compare the grid with the VSC's internal angle, a PI-based loop filter to minimize error, and a voltage-controlled oscillator (VCO) to generate the synchronized angle (θ). This angle enables the transformation of grid voltages and currents into the rotating dq-frame, simplifying the control scheme. Current loop control regulates the VSC's output current using PI controllers in the dq-frame. Decoupling terms counteract cross-coupling between the q and d axes, allowing active and reactive power to be controlled separately. The PI controllers generate reference signals which are transformed back to the abc-frame using θ and fed to a PWM module to generate switching signals.

This model ensures stable grid integration and independent active/reactive power control. It is foundational for renewable energy systems, microgrids, and HVDC applications, offering flexibility and reliability in modern power systems. The VSC blocks with various parts are shown in figure 1.

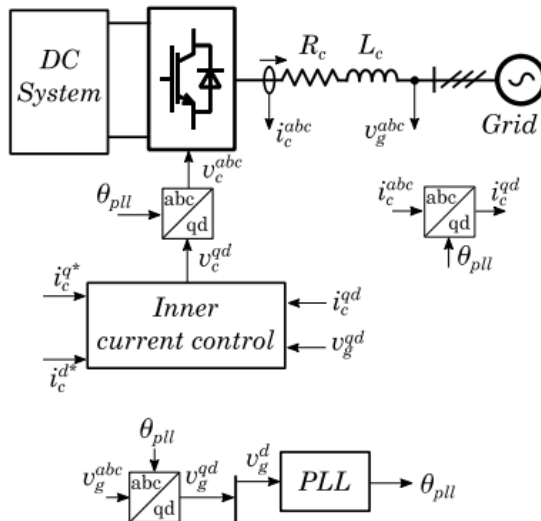


Figure 1: Basic VSC Model with PLL and Current Loop Control

1.2 Model Parameters for the Implementation

1.2.1 Parameters and Data for the VSC Model

```
1 % System parameters
2 Scn = 5e6; % Nominal power [VA]
3 Ucn = 690; % Nominal AC voltage [V]
4 fg = 50; % Base frequency [Hz]
5 Lc = 30.31e-6; % Converter inductance [H]
6 Rc = 0.95e-3; % Converter resistance [ohm]
7 Vpeak = Ucn*sqrt(2)/sqrt(3); % Peak voltage [V]
```

1.2.2 PLL Tuning Parameters

```
1 % PLL parameters
2 xi_pll = 0.707; % Damping ratio (sqrt(2)/2)
3 omega_pll = 2*pi*50; % Natural frequency [rad/s]
4 kp_pll = xi_pll*2*omega_pll/Vpeak; % Proportional gain
5 tau_pll = 2*xi_pll/omega_pll; % Time constant [s]
6 ki_pll = kp_pll/tau_pll; % Integral gain
```

1.2.3 Current Control Parameters

```
1 % Current controller parameters
2 tauc = 1e-3; % Time constant [s]
3 kpc = Lc/tauc; % Proportional gain
4 kic = Rc/tauc; % Integral gain
```

1.2.4 Clarke Transformation ($abc \rightarrow \alpha\beta 0$)

- Converts three-phase quantities to stationary reference frame:

$$\begin{bmatrix} v_\alpha \\ v_\beta \\ v_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix}$$

- Eliminates redundant information in balanced systems
- Preserves power equivalence: $P_{abc} = P_{\alpha\beta 0}$
- v_0 component vanishes in balanced conditions

1.2.5 Park Transformation and Inverse Park Transformation ($\alpha\beta \rightarrow dq$ and $dq \leftarrow \alpha\beta$)

- Rotates coordinates to synchronous reference frame:

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix}$$

- Aligns with grid voltage vector ($\theta = \omega t$)
- Converts AC quantities to DC values in steady state
- Enables:
 - Independent control of active (v_d) and reactive (v_q) power
 - Simplified PI controller implementation
 - Frequency/phase synchronization via PLL

The Inverse Park Transformation changes the parameters from qd0 frame back to abc frame. The park inverse model is shown in figure 2.

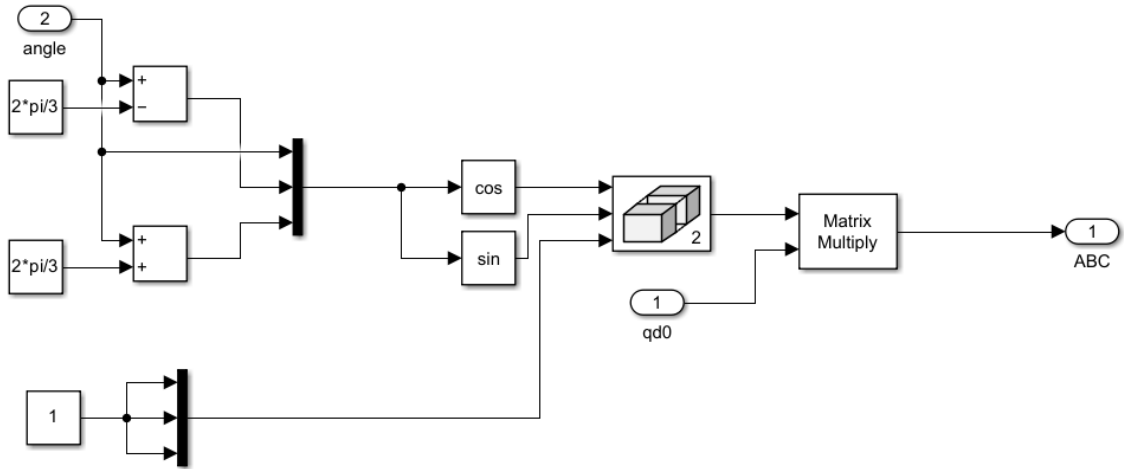


Figure 2: Park Inverse Model

1.2.6 Three-Phase Voltage Source

- Generates balanced three-phase AC voltages:

$$\begin{aligned} v_a(t) &= V_m \cdot \sin(\omega t) \\ v_b(t) &= V_m \cdot \sin\left(\omega t - \frac{2\pi}{3}\right) \\ v_c(t) &= V_m \cdot \sin\left(\omega t + \frac{2\pi}{3}\right) \end{aligned}$$

- Key parameters:

- Amplitude: $V_m = \sqrt{2} \cdot V_{\text{rms}}$
- Frequency: $\omega = 2 \cdot \pi \cdot f$ (typically 50/60 Hz)
- 120° phase displacement between phases. The implementation is shown in figures 3 and 4 and produces a typical three-phase, 690 V, 50 Hz response.

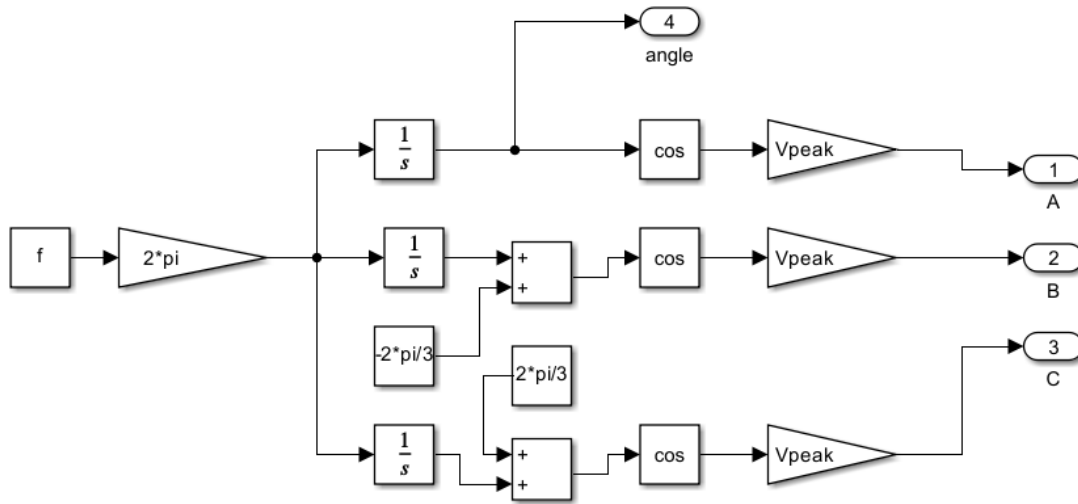


Figure 3: Three phase voltage source generation

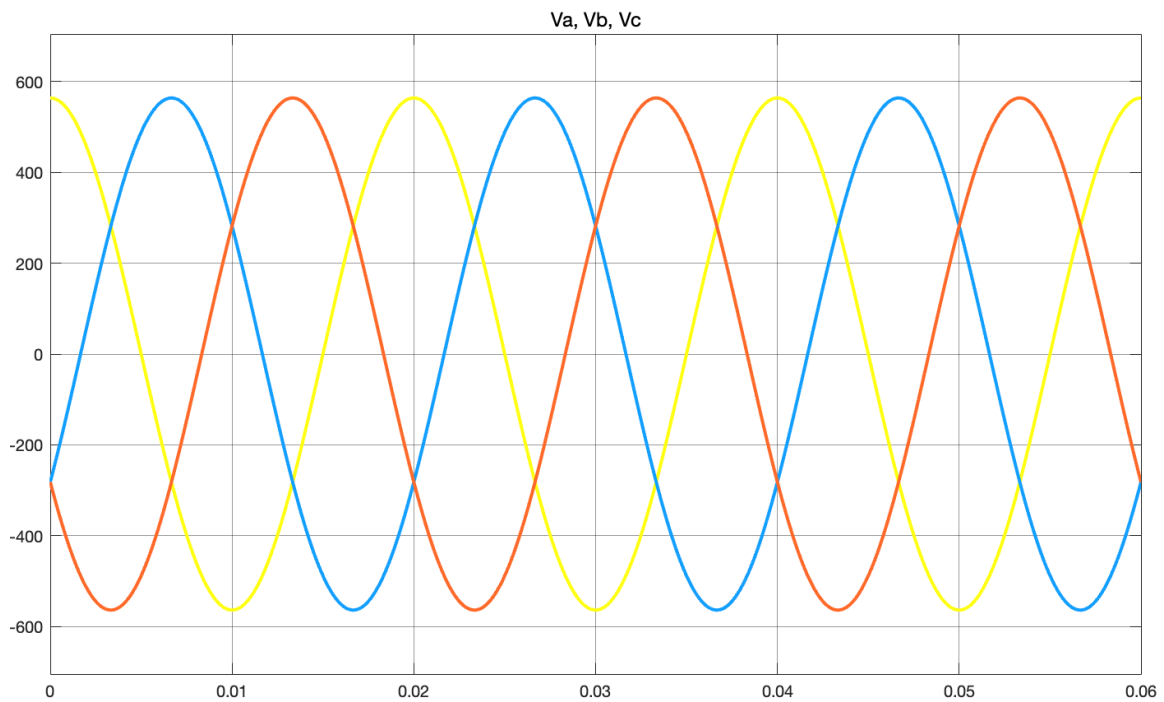


Figure 4: Three phase voltage source validation

1.3 PLL implementation

A Phase-Locked Loop (PLL) is utilized to estimate the angle and angular velocity of an electrical network, enabling the determination of the grid voltage angle. Accurately latching to the grid angle is essential for effective converter control. The model of the PLL implementation is illustrated in figure 5.

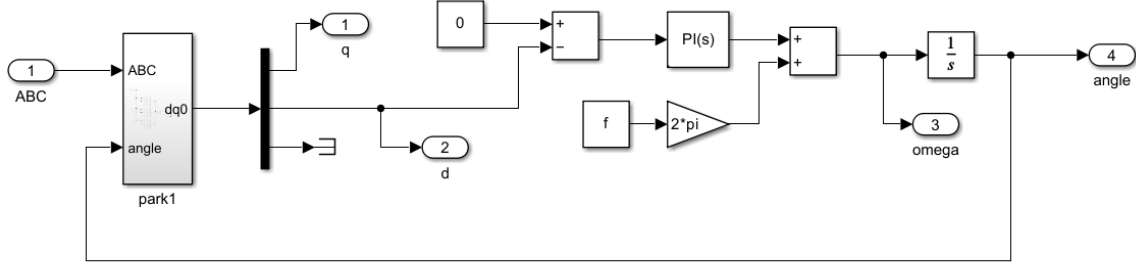


Figure 5: PLL Simulink model

The PLL PI controller is expressed as in equation 1, where τ_{PLL} is the time constant of the PLL.

$$K_f(s) = K_p \left(\frac{\frac{1}{\tau_{\text{PLL}}} + s}{s} \right) \quad (1)$$

By using the controller parameters using the equation 2 and 3, K_p and τ_{PLL} can be computed as 4

$$\omega_n = \sqrt{\frac{K_p V_{\text{peak}}}{\tau_{\text{PLL}}}} \quad (2)$$

$$\xi = \frac{\sqrt{\tau_{\text{PLL}} K_p V_{\text{peak}}}}{2} \quad (3)$$

$$K_p = \frac{2\xi\omega_n}{V_{\text{peak}}}; \quad K_i = \frac{K_p}{\tau_{\text{PLL}}}; \quad \tau_{\text{PLL}} = \frac{2\xi}{\omega_n} \quad (4)$$

The graph shows the increasing angle, as illustrated in figure 6. The θ_{real} (green line) represents the actual phase angle of the grid. It increases over time because the system has a constant rotational speed. The θ_{PLL} (orange line) is the estimated phase angle from the PLL, used to track the real angle. We can observe that the orange and green lines overlap, which means the PLL has successfully locked onto the actual phase.

Meanwhile, the w_{PLL} (blue line) represents the estimated angular velocity of the grid. Its constant value indicates that the system is operating at a steady frequency. This confirms that the PLL is working successfully.

1.3.1 PLL implementation validation

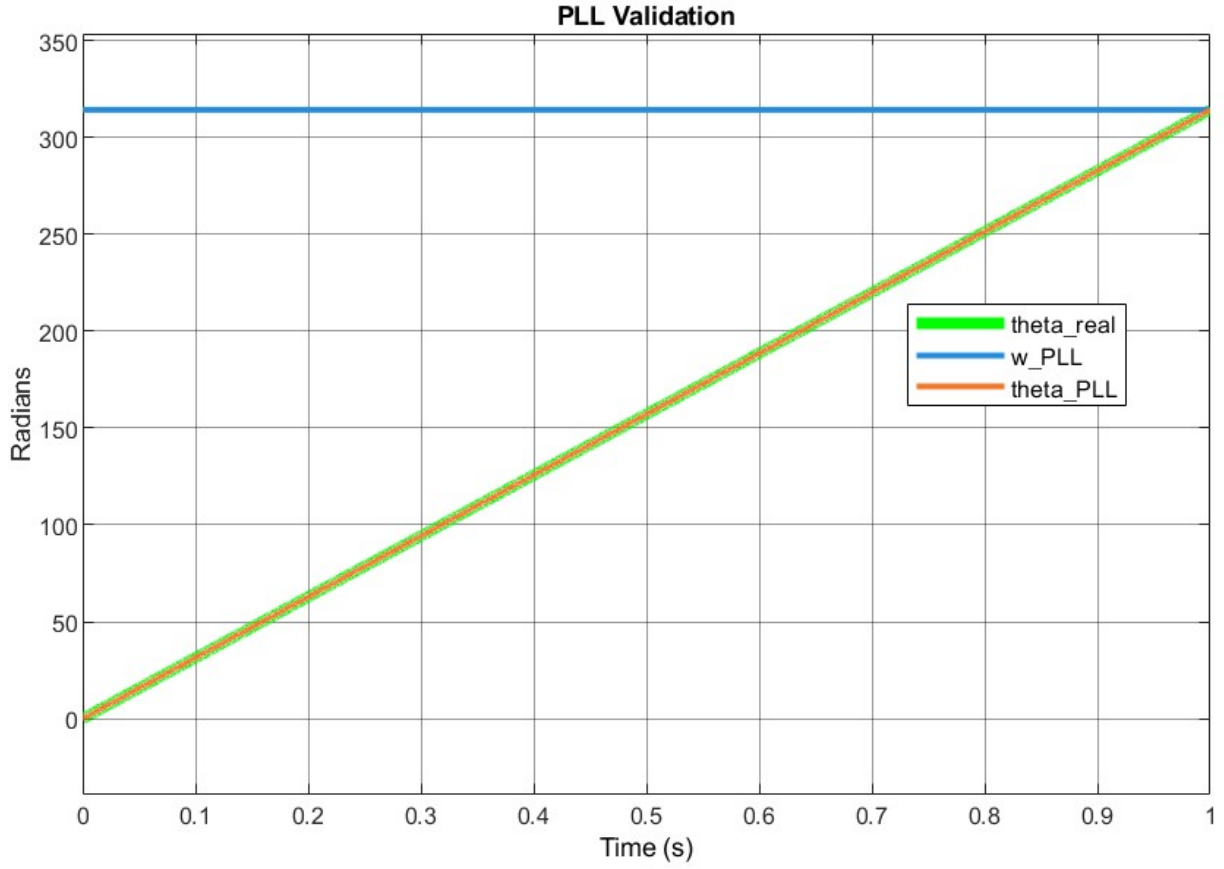


Figure 6: PLL angle simulation

1.4 Current Loop Implementation

The current control loop is responsible for comparing the reference and actual current values in the dq frame. According to [1] and [2], the optimal design of the current loop involves using a PI controller, which approximately cancels the pole introduced by the filter. The PI current controller is expressed as in equation 5.

$$k(s) = \frac{k_{pi} \cdot s + k_{ii}}{s} \quad (5)$$

Constants of the PI controller mainly depend on the filter parameters, as shown in equation 6.

$$k_{pi} = \frac{\sum L_f}{\tau_i}; \quad k_{ii} = \frac{R_f}{\tau_i}. \quad (6)$$

τ_i is a design parameter that directly influences the performance of the current loop. A smaller value results in a faster response; however, it must be sufficiently large to ensure that $\frac{1}{\tau_i}$ remains small relative to the VSC switching frequency (in radians per second). Typically, τ_i is chosen to be at least ten times faster than the converter switching frequency. Figure 7 shows the Current Loop model and the decoupling of the dq components.

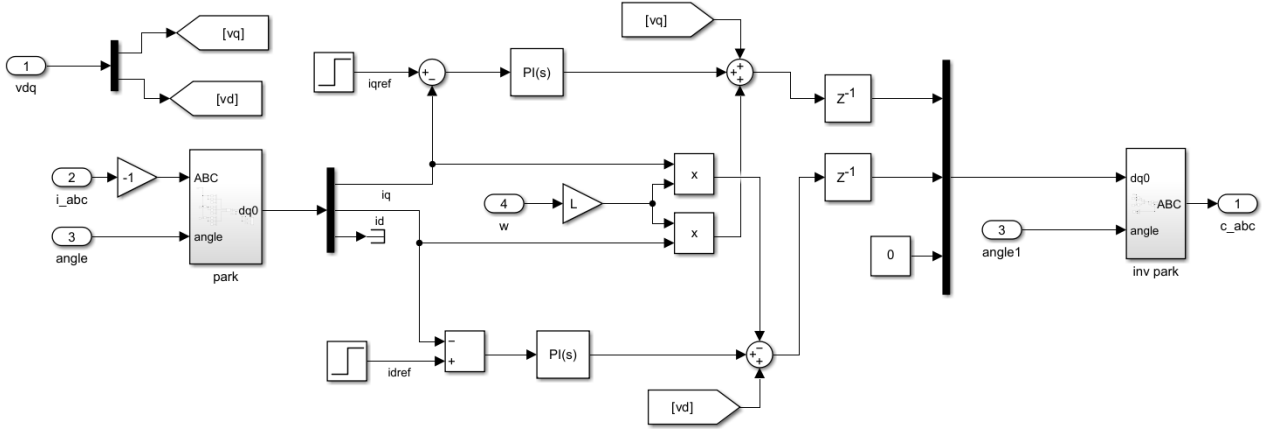


Figure 7: Current Loop Model

1.5 Presentation of a time-domain simulation

The validation for the basic VSC model is performed by simulating all completed loops connected together as shown in figure 8.

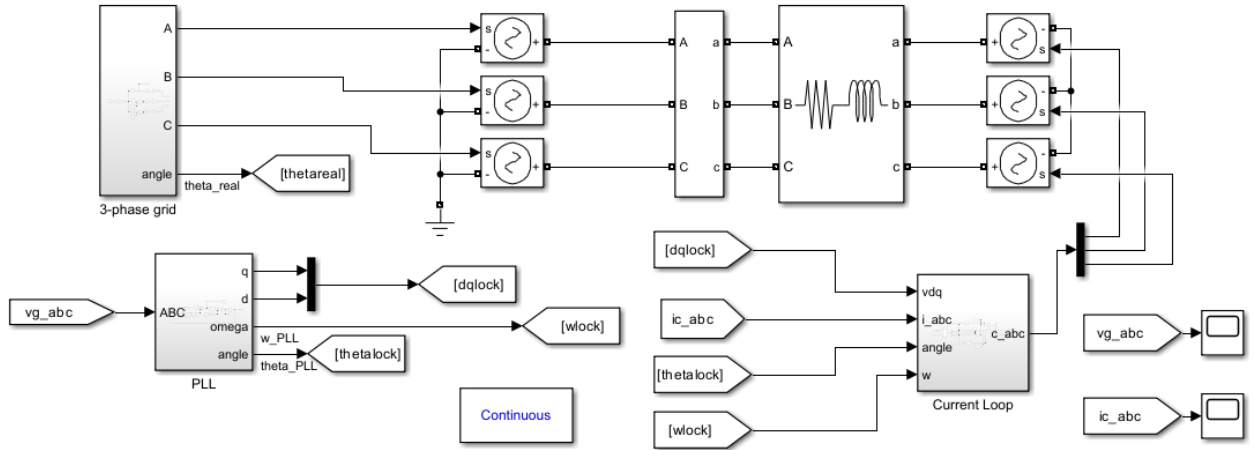


Figure 8: Time-Domain Simulation Current

The current references both start at 0 and increase to 2 kA and 1 kA for i_d and i_q , respectively. This response of the model is shown in figure 9.

The two currents follow their respective references precisely with a good response time, so the model is confirmed to function properly. To find the time constant τ_C , the first order response is examined in Figure 10. 63.2% of the reference current 2000 A is 1264 A, so the shown point is as close as possible and gives a time constant of 1 ms.

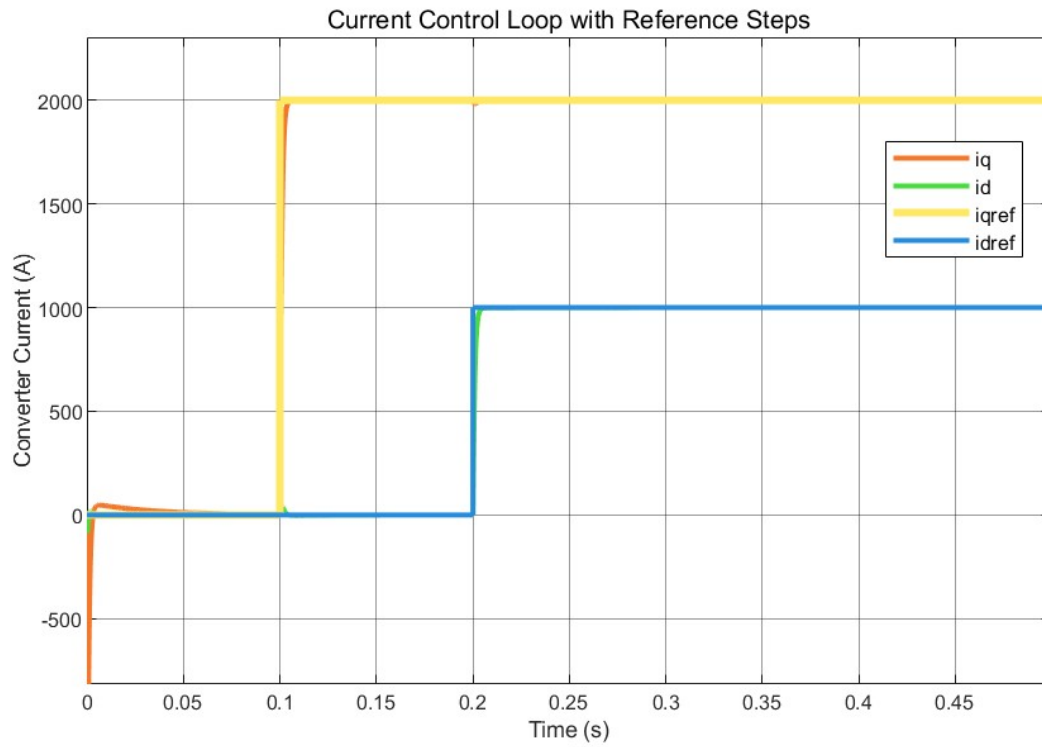


Figure 9: Time-Domain Simulation Current

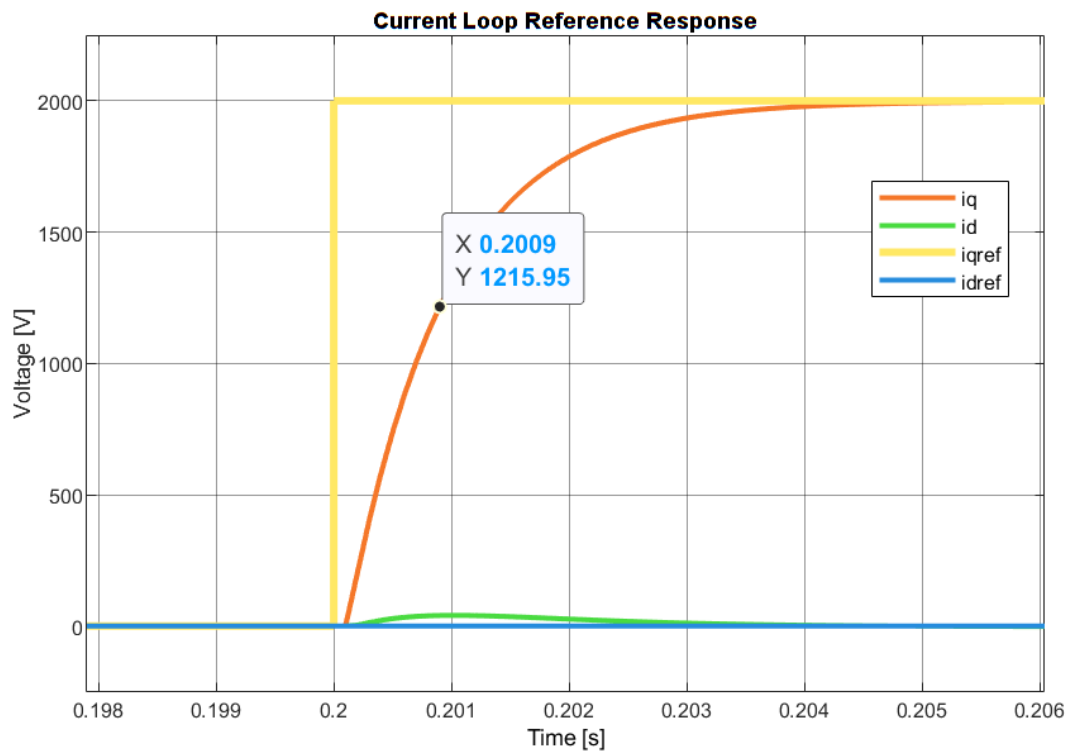


Figure 10: Current Response Time Constant

2 Task 1.2 - Adding Power Control Loops

Figure 11 shows the power loop control, considering the direct current reference calculation and the cascaded PI control.

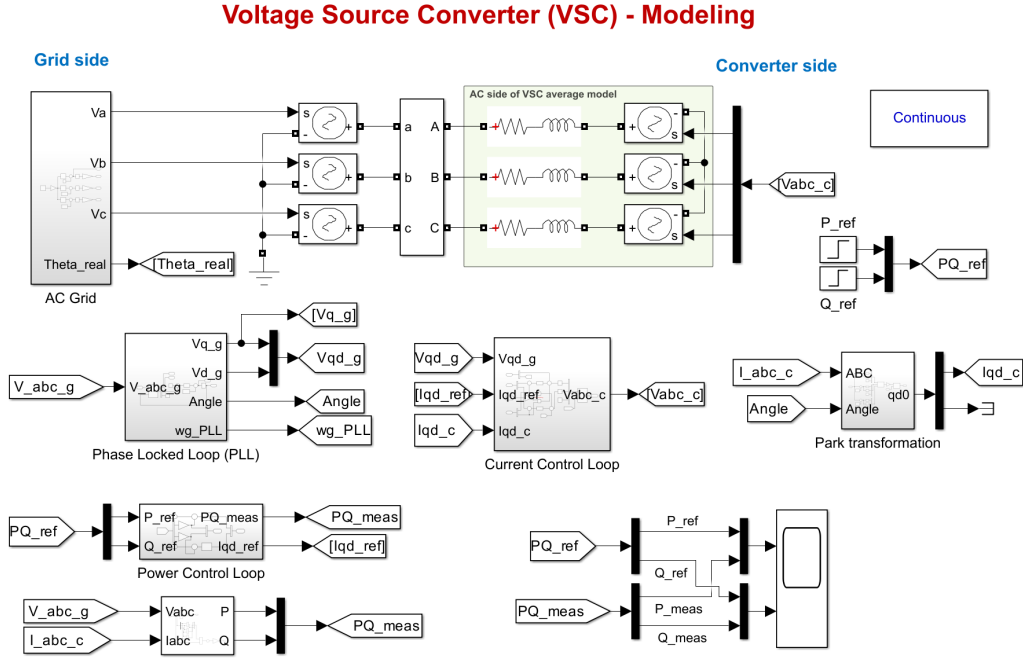
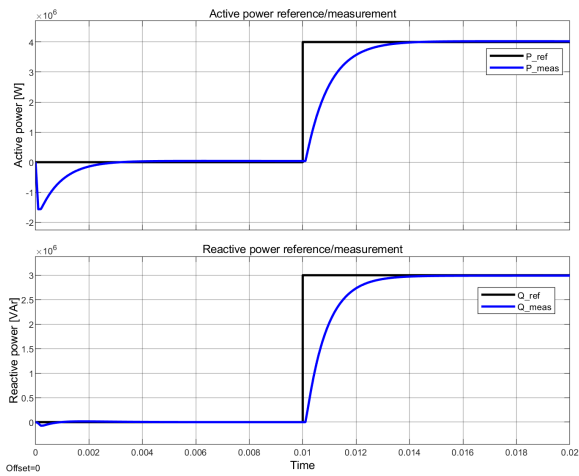


Figure 11: Voltage source converter model with power control loop

2.1 Direct current reference calculation

Power control loops can be implemented using only the direct current reference calculation by applying the equation 7. Note that the measured voltage V_{zq} is equal to V_{peak} only for balanced grids, meaning the system operates under ideal conditions, with no imbalances or distortions in the voltage waveforms. For an unbalanced grid, those values are not equal due to the effects of waveform distortion on the voltage.

$$I_q^* = \frac{P^* \cdot 2}{3 \cdot V_{zq}} \quad I_d^* = \frac{Q^* \cdot 2}{3 \cdot V_{zq}} \quad V_{zq} = V_{peak} \quad (7)$$



(a) P and Q calculations using the constant value

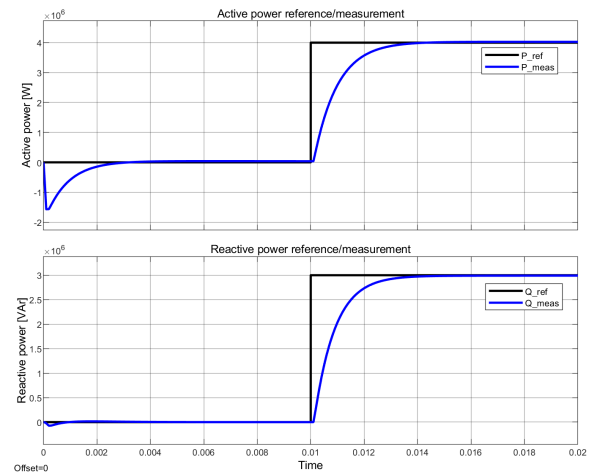

 (b) P and Q calculations using the measured voltage value V_{zq}

Figure 12: Active and reactive power calculations using the constant or measured value

The results of the different calculations, considering the constant value and measured voltage, are shown in figure 12a and figure 12b, respectively. There is no difference between these two values, as the grid is balanced.

2.2 Cascaded PI control

Another alternative for building the power control loop is to create a cascaded PI control according to the schematic presented in figure 13. It is important to note that the time response can be adjusted by modifying the time constant τ_p .

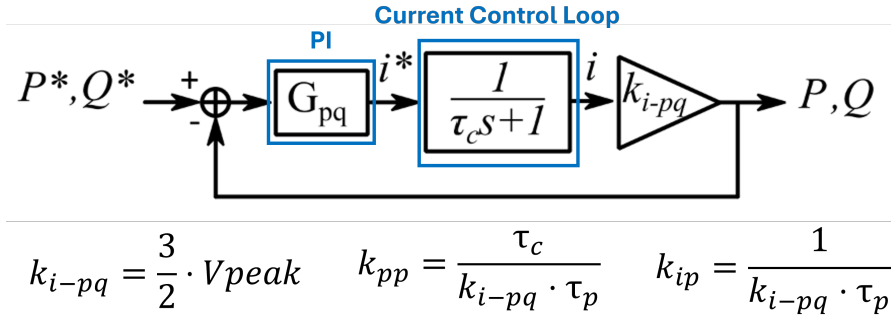


Figure 13: Cascade PI control

By increasing the active power (4 MW at 0.01 seconds) and reactive power (3 MVar at 0.01 seconds), different dynamic responses can be observed in the system. Figures 14, 15, and 16 illustrate the time responses corresponding to 5 ms, 10 ms, and 100 ms, respectively. These variations highlight the impact of different time constants on the system's transient behavior, demonstrating how response speed can be influenced by power adjustments. Each figure shows a point that represents the time $t = \tau_p$, at which the output has reached 63.2% of its steady-state value.

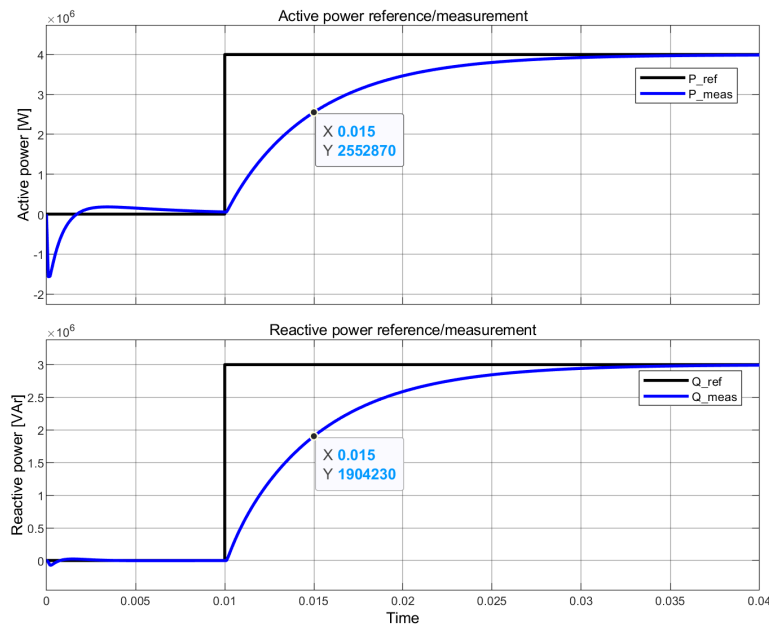


Figure 14: Time response of 5 ms

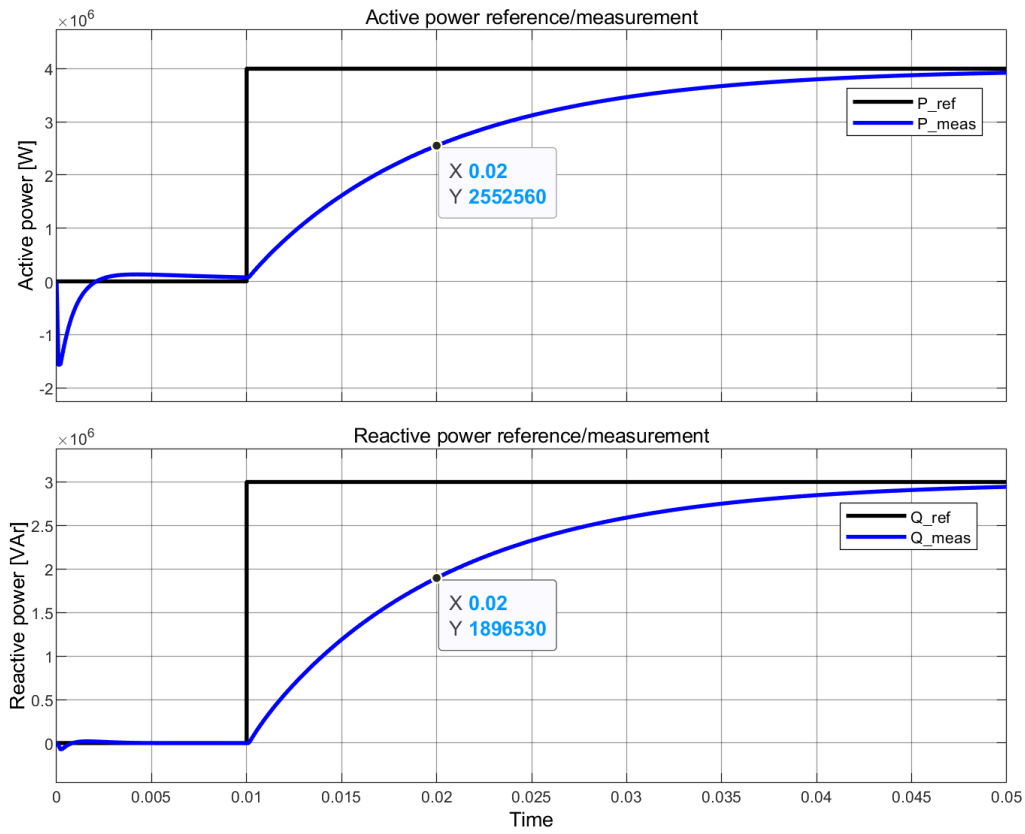


Figure 15: Time response of 10 ms

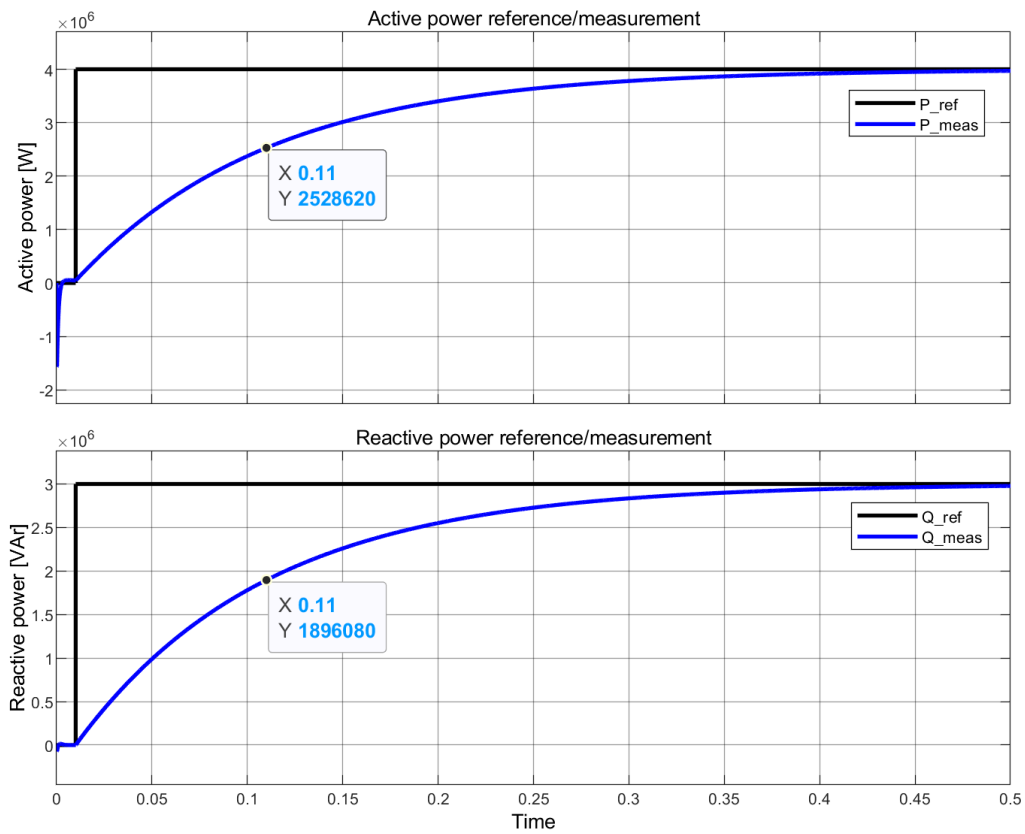


Figure 16: Time response of 100 ms

2.3 Discussion of first-order behavior in case of cascaded control

The previous section 2.2 shows the cascaded controller diagram and the first-order response that it exhibits. k_{pp} and k_{ip} are selected such that G_{pq} cancels the pole from the current control loop, allowing the P/P* model in figure 13 to exhibit a first-order characteristic such as in figures 14, 15, and 16.

However, this model of the power loop is only valid when τ_p is much larger than τ_c . For the cases of τ_p equal to 5, 50, and 100 ms, the current loop has time for its pole at $\frac{1}{\tau_c s + 1}$ to be cancelled, and the whole system becomes first order. When this is not the case, the system exhibits its higher-order dynamics such as in figure 17. This interaction is discussed further in the next section.

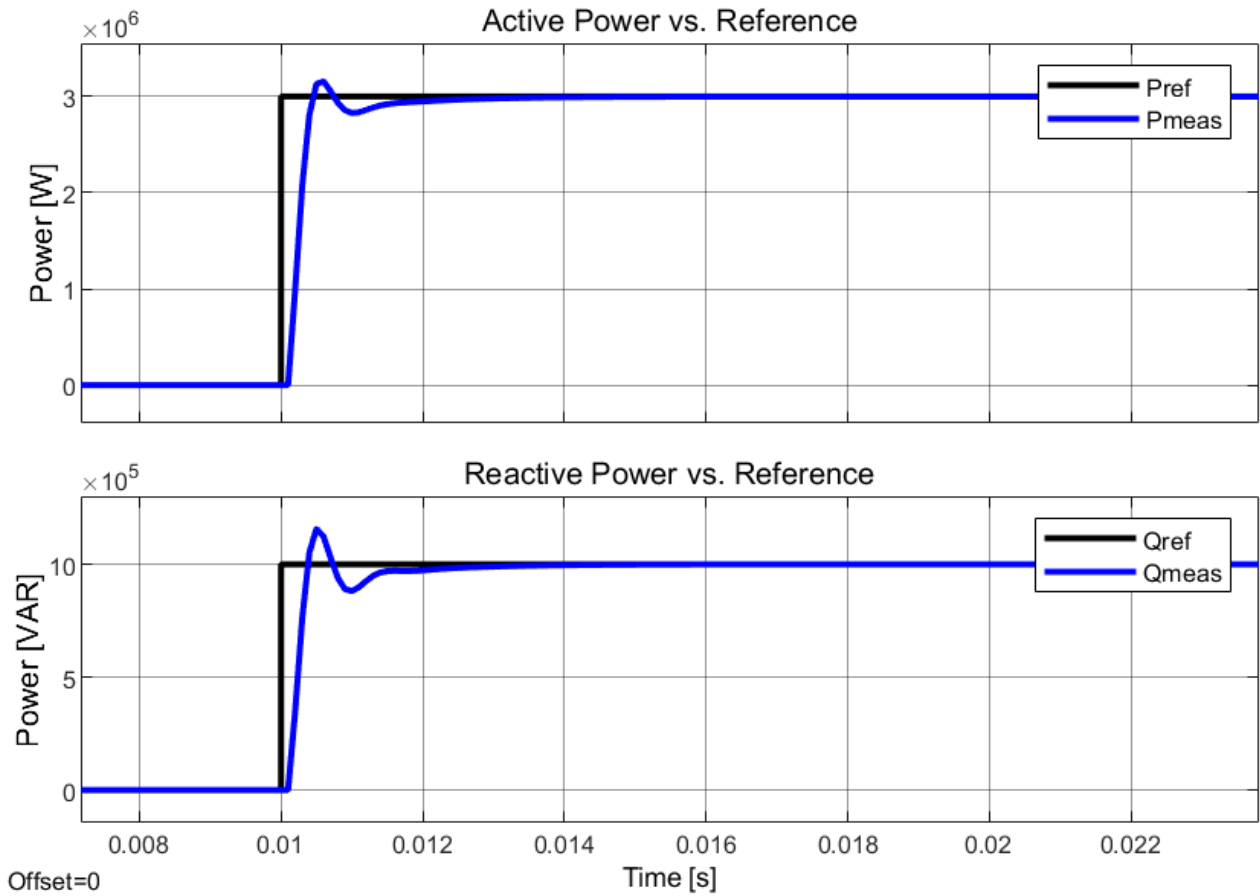


Figure 17: Power response for $\tau_p = 0.3\tau_c$

2.4 Comparison of both control implementations

Using the direct current reference calculation approach, the current reference is obtained directly from the power equation 7. Since this approach does not use an additional PI controller, the response time is limited only by the current control dynamics. There is no additional lag introduced by the outer power loop, as the reference current is calculated in real-time from the power reference. Hence, the response using this approach is faster because the power dynamics follow the current loop directly.

Using the cascade PI control approach, the power reference and actual power measurement go through a PI controller before generating the current reference. In this structure, the outer loop (power loop) must be at least one decade slower than the inner loops to exhibit a first order response.

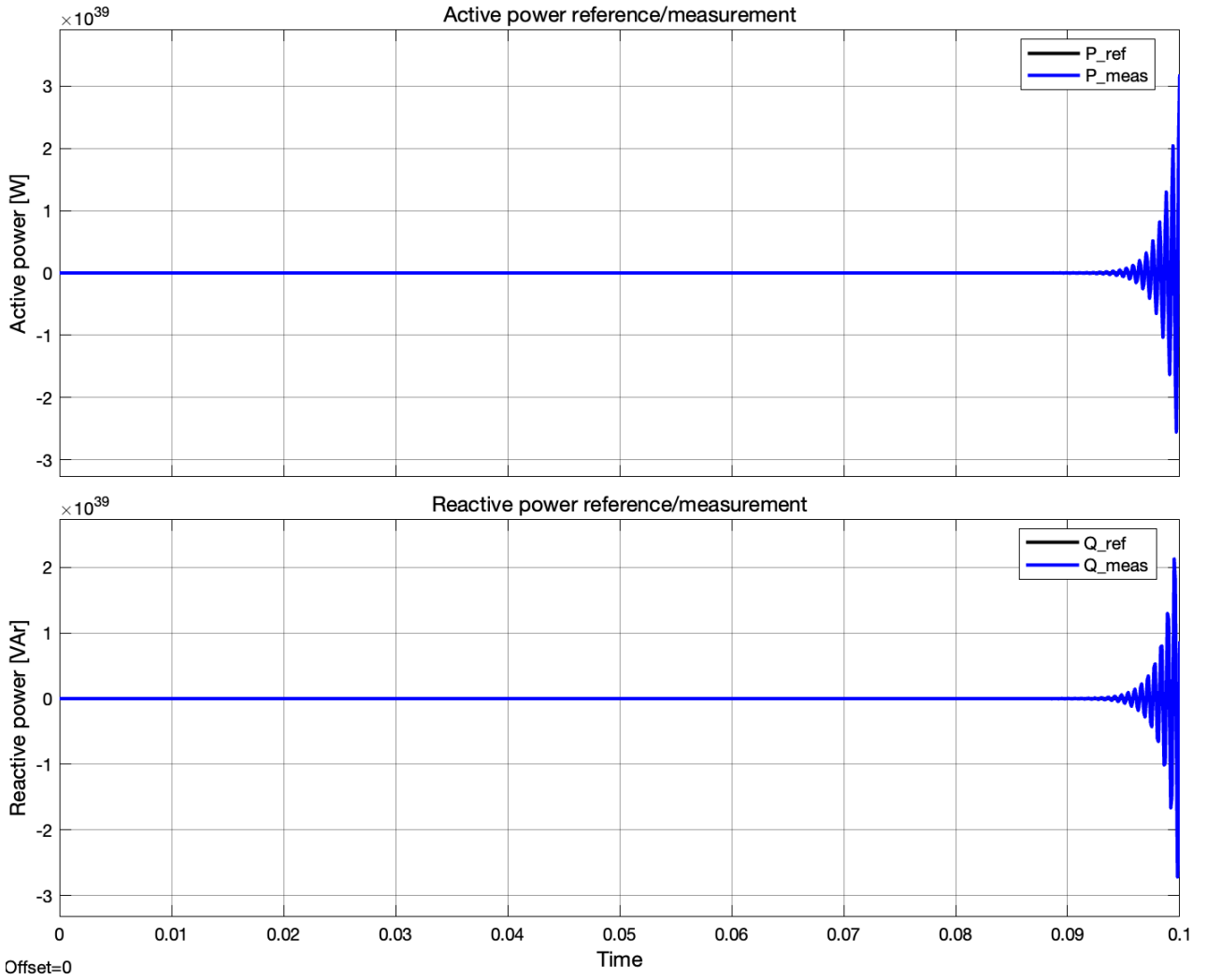


Figure 18: Time response of power control loop faster than current control loop

By setting the time response of the power control loop in the cascade PI control approach faster than the time response of the current control loop, an unstable response is observed in figure 18. This instability occurs because the current control loop is responsible for regulating the actual current based on the reference in the dq frame, ensuring the power injection. Meanwhile, the outer power control loop takes the measured power—derived from the actual current in the dq frame—and compares it with the power reference to generate a power error.

If the power loop reacts too quickly, it may generate large, rapid changes in the current reference before the inner current loop can effectively track them. This mismatch causes oscillations, as the power loop continuously updates the reference without allowing the current loop to settle, leading to an unstable system response.

Therefore, to ensure stability, the current control loop must be significantly faster than the power control loop.

2.5 Validation of steady state results (Voltage, Current, and Power) with MATLAB script

The validation of the Voltage Source Converter (VSC) model can be done by comparing the steady-state voltages, currents, and power obtained from the Simulink model with analytical calculations from a MATLAB script.

First, the results from the previous section (Figure 14) are used to support the assumption that both powers are equal to their respective references in steady-state. The constant P and Q are used with the grid voltage to calculate the peak values for the converter voltage and the current, as shown in figure 19 below:

VSC Verification

```
Pref = 3e6;  
Qref = 1e6;  
Vg = U/sqrt(3); % grid rms  
Ig = (Pref - 1j*Qref)/3/Vg; % single phase current  
Ig_peak = abs(Ig)*sqrt(2)  
  
Z = Rc + 1j*2*pi*f*L; % R + X  
Vc = Vg + Ig*Z;  
Vc_peak = abs(Vc)*sqrt(2)
```

Ig_peak = 3.7420e+03

Vc_peak = 578.9460

Figure 19: Matlab Script of VSC Voltage and Current

The calculated peak voltage can be validated with the q component of the voltage in the simulation, since these are equivalent at steady state operation. The converter V_q is monitored with a Simulink Display block in figure 20.

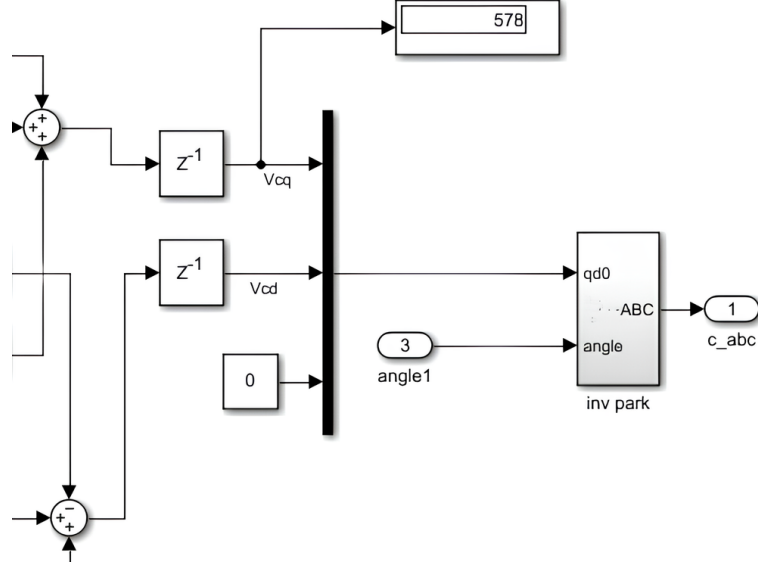


Figure 20: Q component

The calculated peak current must be validated with both the steady-state I_d and I_q from the simulation, which are shown in figure 21 below:

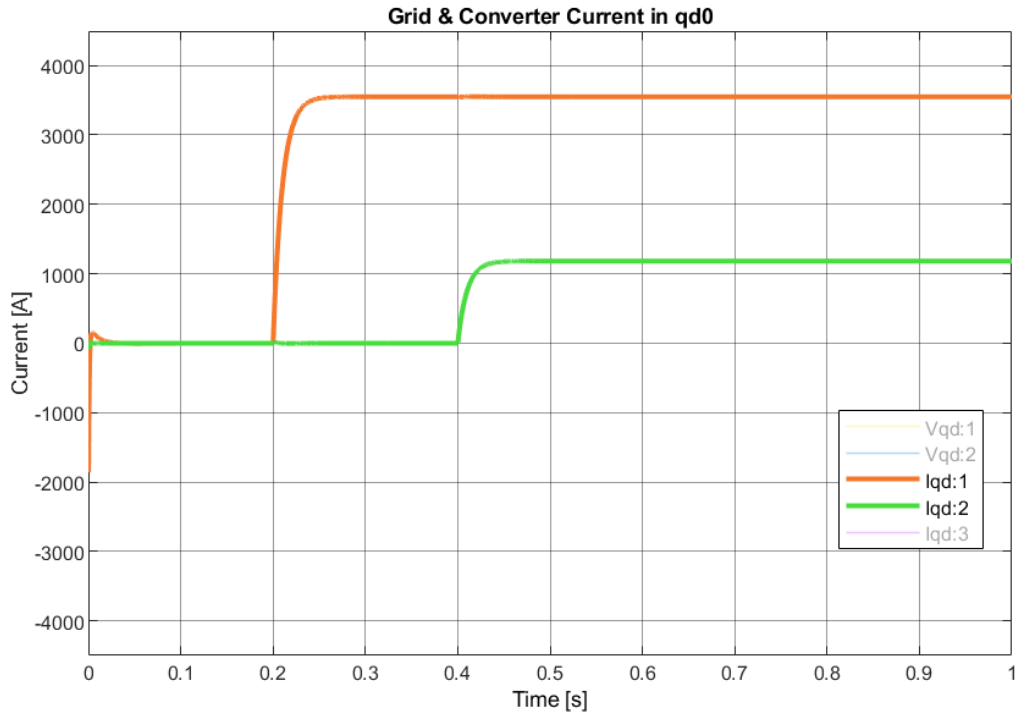


Figure 21: Converter Current under Power Reference Steps

The steady state currents are 3556 and 1182 A for I_q and I_d , respectively. The magnitude of these currents is 3.74 kA, which is indeed equal to the calculated peak current.

3 Task 1.3 - Adding Grid Impedance

3.1 X/R Ratio

This parameter determines the behavior of fault currents, including their peak magnitude, waveform asymmetry, and decay rate. The X/R ratio, defined as the ratio of reactance to resistance in a circuit, reflects the influence of the DC component of the fault current, which is assumed to decay exponentially [3].

In power systems, where inductive components, such as transformers, generators, and transmission lines dominate, the X/R ratio has a significant impact on the transient response of the system during fault conditions. A higher X/R ratio results in a slower decay of the DC offset in the fault current, leading to greater asymmetry and increased peak currents.

3.2 Short Circuit Ratio (SCR)

The SCR is a fundamental metric in power systems used to quantify the strength of the electrical grid at a specific connection point. It is defined as the ratio of the short circuit apparent power (SCMVA) from a three-phase-to-ground (3LG) fault at a given location in the power system to the rated capacity of the inverter-based resource connected to that location. This metric is commonly applied at connection points that involve technologies such as HVDC systems or wind farms, as it helps assess their impact on the system [4].

A low SCR area (referred to as a “weak system”) indicates a high sensitivity of voltage (both in magnitude and phase angle) to changes in active and reactive power injections or consumptions. High SCR (referred to as a “stiff system”) exhibits a low sensitivity and is predominantly unaffected by such changes. A strong grid has low impedance, meaning it can supply or absorb large amounts of current with minimal voltage fluctuation during disturbance. Some representative SCR values are shown below, according to [5].

Table 1: SCR Classification

SCR	Classification
$\text{SCR} > 3$	Strong system
$2 < \text{SCR} < 3$	Weak AC system
$\text{SCR} < 2$	Very weak AC system

To account for different systems connected to the voltage source converter, Thevenin impedance and SCR are employed to develop the appropriate implementation using equations 8 - 10. However, it is important to note that using Thevenin impedance is an approximation, as this equivalent cannot capture the varying saturation states of power converters. Therefore, this parameter must be calculated correctly [6].

$$X_{\text{ratio}} = \frac{X_s}{R_s} \quad \text{SCR} = \frac{S_{\text{cc MVA}}}{\text{MW}_{\text{sources}}} = \frac{V_{s_LL}^2}{Z_s \cdot \text{MW}_{\text{sources}}} \quad (8)$$

$$sfdZ_s = \sqrt{R^2 + X^2} \quad R_s = \sqrt{Z_s^2 - X^2} = \frac{Z_s}{\sqrt{1 + X_{\text{ratio}}^2}} \quad (9)$$

$$X_s = X_{\text{ratio}} \cdot R_s \quad (10)$$

Where:

- $S_{\text{cc MVA}}$ = Short circuit 3LG at the connection point
- $\text{MW}_{\text{sources}}$ = Nominal rated power of the inverter-based resource at the connection point
- V_{s_LL} = Voltage line-to-line
- Z_s = Thevenin impedance
- R_s = Thevenin resistance
- X_s = Thevenin reactance

Figure 22 shows the model considering the grid impedance. In the following sections, different values of grid impedance will be considered in order to observe the system's response behavior.

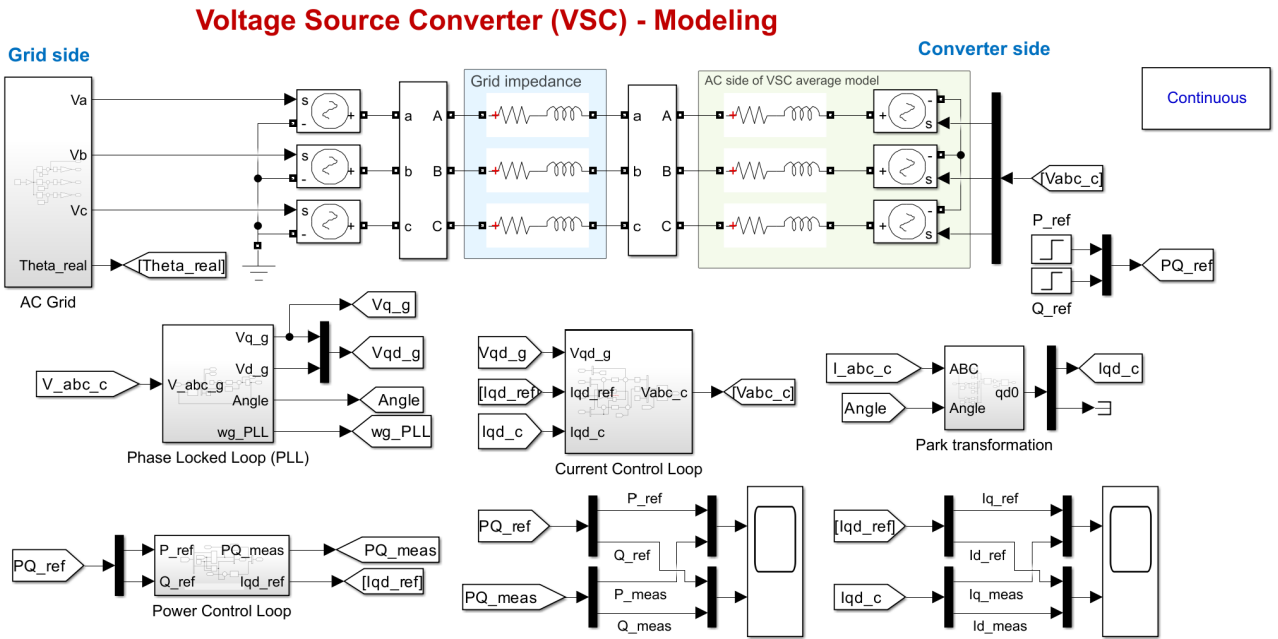
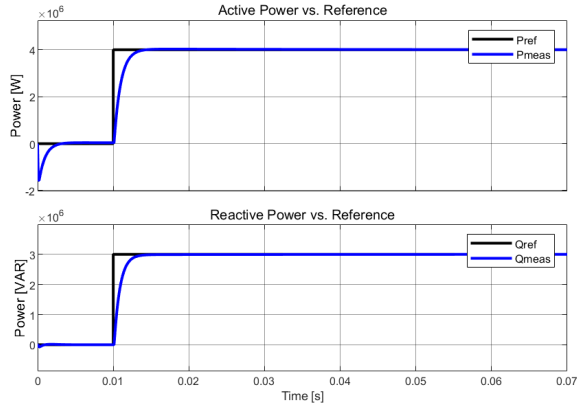


Figure 22: Voltage source converter model with grid impedance

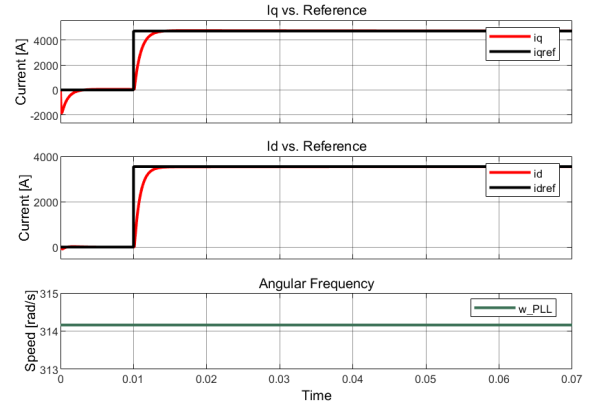
3.3 Comparison of results with direct current reference calculation approach for power control

The analysis of the VSC using direct reference calculation in the power loop under different Short Circuit Ratio (SCR) conditions is shown in the following figures.

3.3.1 Case 1: Without Grid Impedance



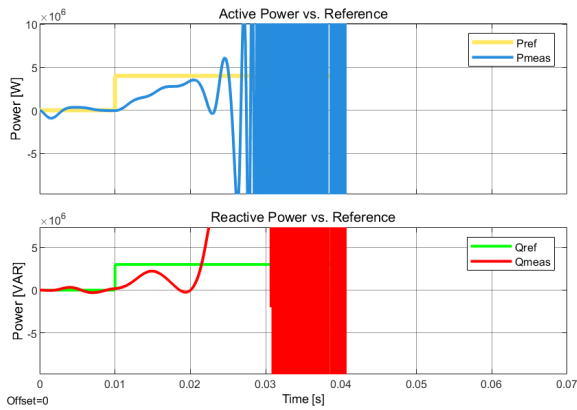
(a) P and Q Response



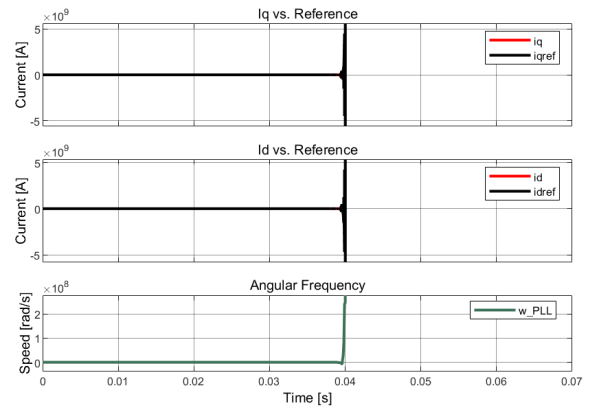
(b) Current and PLL Response

Figure 23: Dynamic response results of powers, currents and PLL frequency without grid impedance

3.3.2 Case 2: with Grid Impedance

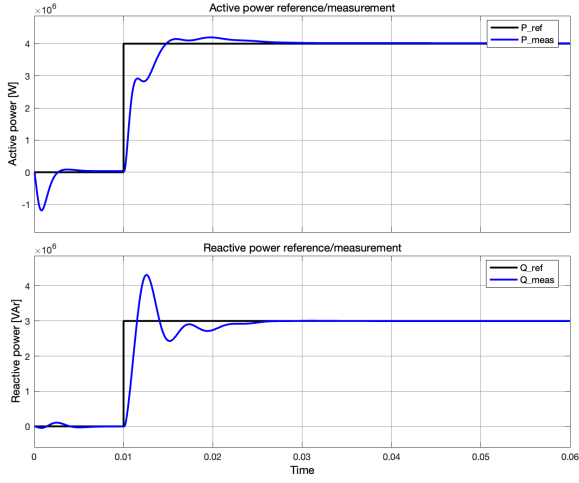


(a) P and Q Response

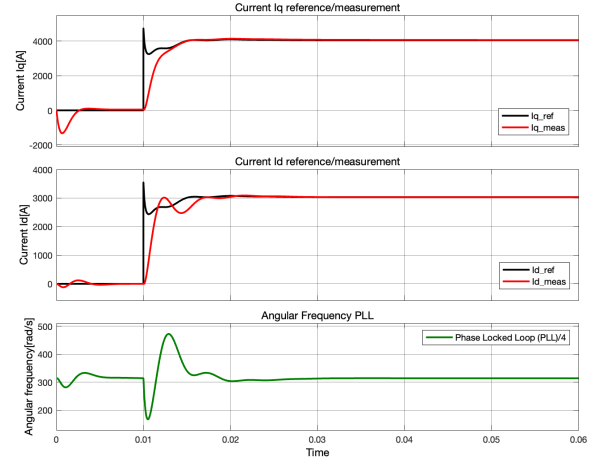


(b) Current and PLL Response

Figure 24: Dynamic response results of powers, currents and PLL frequency for SCR = 1

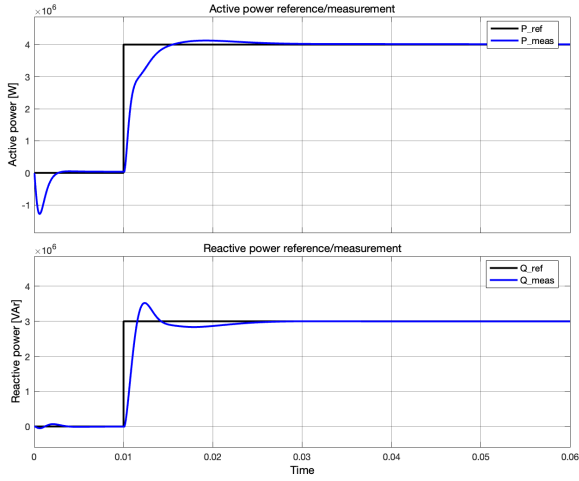


(a) P and Q Response

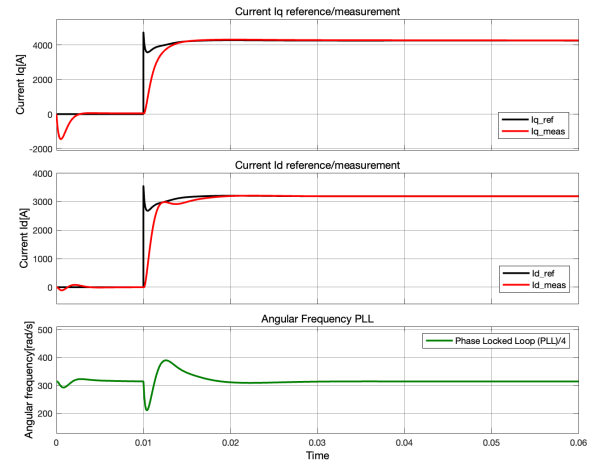


(b) Current and PLL Response

Figure 25: Dynamic response results of powers, currents and PLL frequency for $SCR = 3$



(a) P and Q Response



(b) Current and PLL Response

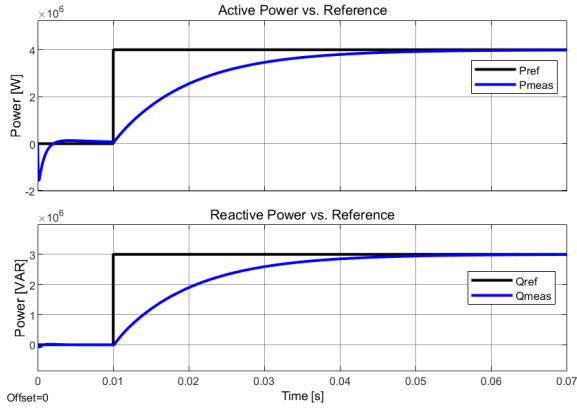
Figure 26: Dynamic response results of powers, currents and PLL frequency for $SCR = 5$

The direct reference calculation approach offers faster system response compared to cascade control, but it comes with a downside—an increased overshoot. This is problematic because the overshoot could push the system beyond its current limit, potentially damaging the voltage source converter (VSC). Introducing the grid impedance caused more dynamics in the PLL response, since the grid's angle is now separated from that of the converter by the inductance. The absence of PLL dynamics in the no-impedance case can be seen in figure 23b. In weak grids (low SCR, high grid impedance), the system becomes more sensitive to disturbances, as seen in figures 24a and 24b. This increased sensitivity makes the overshoot worse, causing instability and would potentially damage a real system.

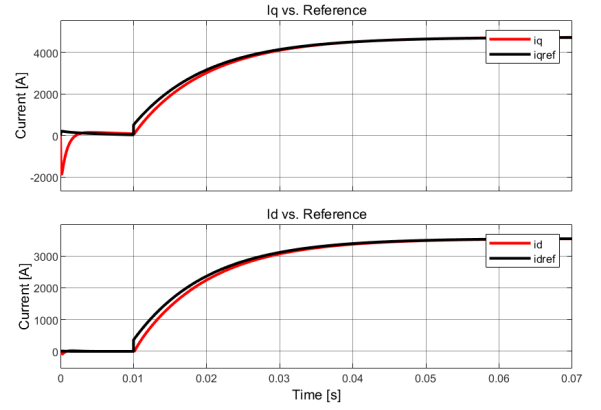
3.4 Comparison of results with cascaded power controller for power control

The analysis of the VSC using cascaded PI control in the power loop under different Short Circuit Ratio (SCR) conditions is shown below.

3.4.1 Case 1: Without Grid Impedance



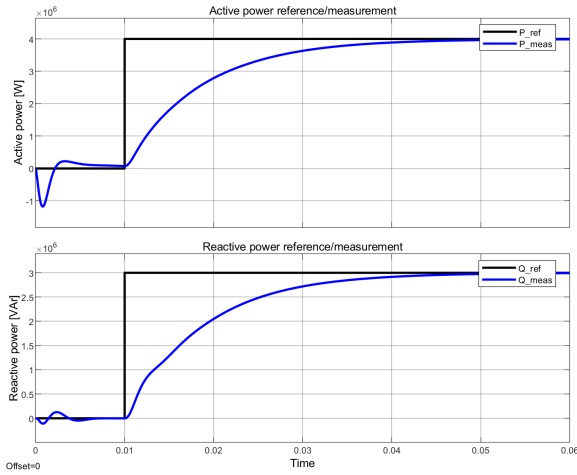
(a) P and Q Response



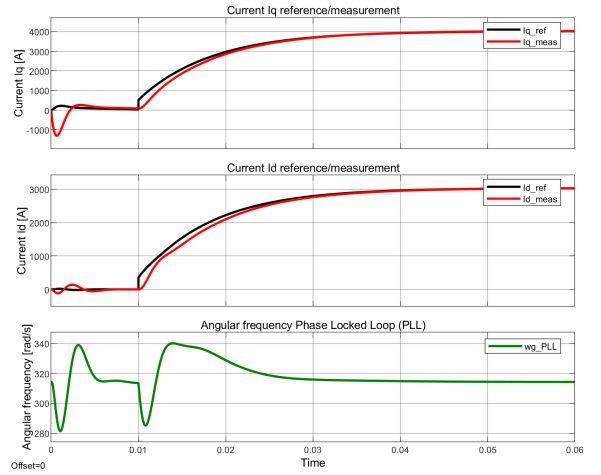
(b) Current and PLL Response

Figure 27: Dynamic response results of powers and currents without grid impedance

3.4.2 Case 2: With Grid Impedance

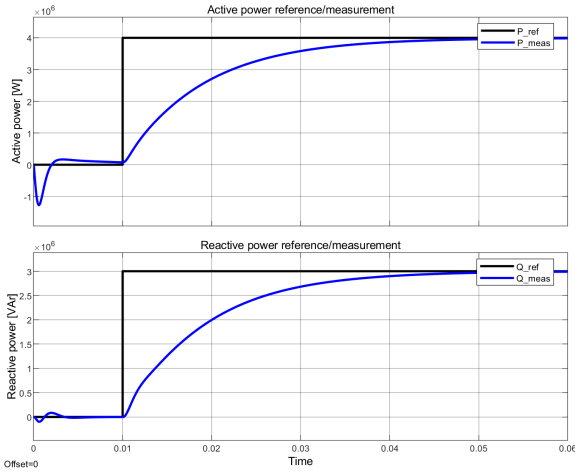


(a) P and Q Response

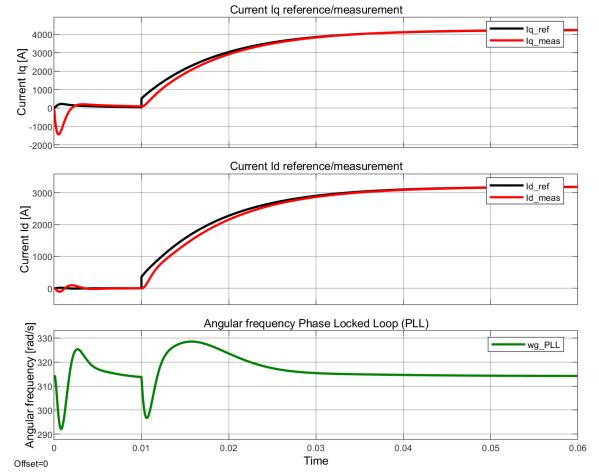


(b) Current and PLL Response

Figure 28: Dynamic response results of powers, currents and PLL frequency for SCR = 3



(a) P and Q Response



(b) Current and PLL Response

 Figure 29: Dynamic response results of powers, currents and PLL frequency for $\text{SCR} = 5$

The results of the cascaded power controller make the slowed current response much more obvious. The reference is now itself the response of a PI controller, so it must reach steady state before the current can. As a result, they move together slowly and without overshoot. Similarly to the direct calculation case, a higher SCR allows the VSC to achieve closer tracking with less ripple. The low SCR case is omitted because it is also an unstable response largely identical to the previous case. The PLL response without grid impedance is also not shown due to the lack of dynamics discussed in 3.3.

3.5 Conclusion

The performance of the Voltage Source Converter (VSC) is significantly influenced by the grid impedance. When integrated with a stronger grid (lower grid impedance), the VSC performs better, transferring more active power from the converter to the grid. This is due to the reduced sensitivity to disturbances, allowing the VSC to operate more efficiently.

However, in weak grid conditions, where the grid impedance is higher, the VSC performance deteriorates. The system becomes more sensitive to disturbances, leading to issues such as overshoot and poor power transfer. In extreme cases, this could cause the VSC to disconnect from the grid to prevent damage, as it struggles to transfer the required power.

One potential solution is to add a voltage controller to the system. This can help regulate the VSC operation more effectively under weak grid conditions by providing more robust control over voltage fluctuations. Additionally, designing a current controller that accounts for real-time grid impedance would allow for better performance by adapting the control strategy to the dynamic grid conditions [7].

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